

Exp no :09

Date :

Using Process Explorer to Identify Suspicious Processes

Description

- Process Explorer is a part of the **Microsoft Sysinternals Suite**. It is a powerful tool used to view detailed information about system processes. It helps investigators and administrators analyze active processes, detect suspicious behavior, monitor CPU and memory usage, and verify process authenticity using digital signatures.

Step-by-Step Process

Step 1: Download and Setup Process Explorer

1. Go to the official Microsoft Sysinternals website: <https://learn.microsoft.com/en-us/sysinternals/downloads/process-explorer>
2. Click **Download Process Explorer**.
3. Extract the downloaded ZIP file to a folder.
4. Right-click procexp64.exe (for 64-bit) or procexp.exe (for 32-bit) → select **Run as Administrator**.

Step 2: Understand the Interface

1. The main window displays all running processes in a **hierarchical tree view**.
 2. Each process shows details such as **PID, CPU usage, memory usage, and company name**.
 3. Color codes represent process states:
 - **Green** — Newly started processes
 - **Red** — Terminated processes
 - **Light Blue** — Processes running under the current user
 - **Pink** — Suspended processes
-

Process Explorer - Sysinternals: www.sysinternals.com [NITHINROYAL\nithi] (Administrator)

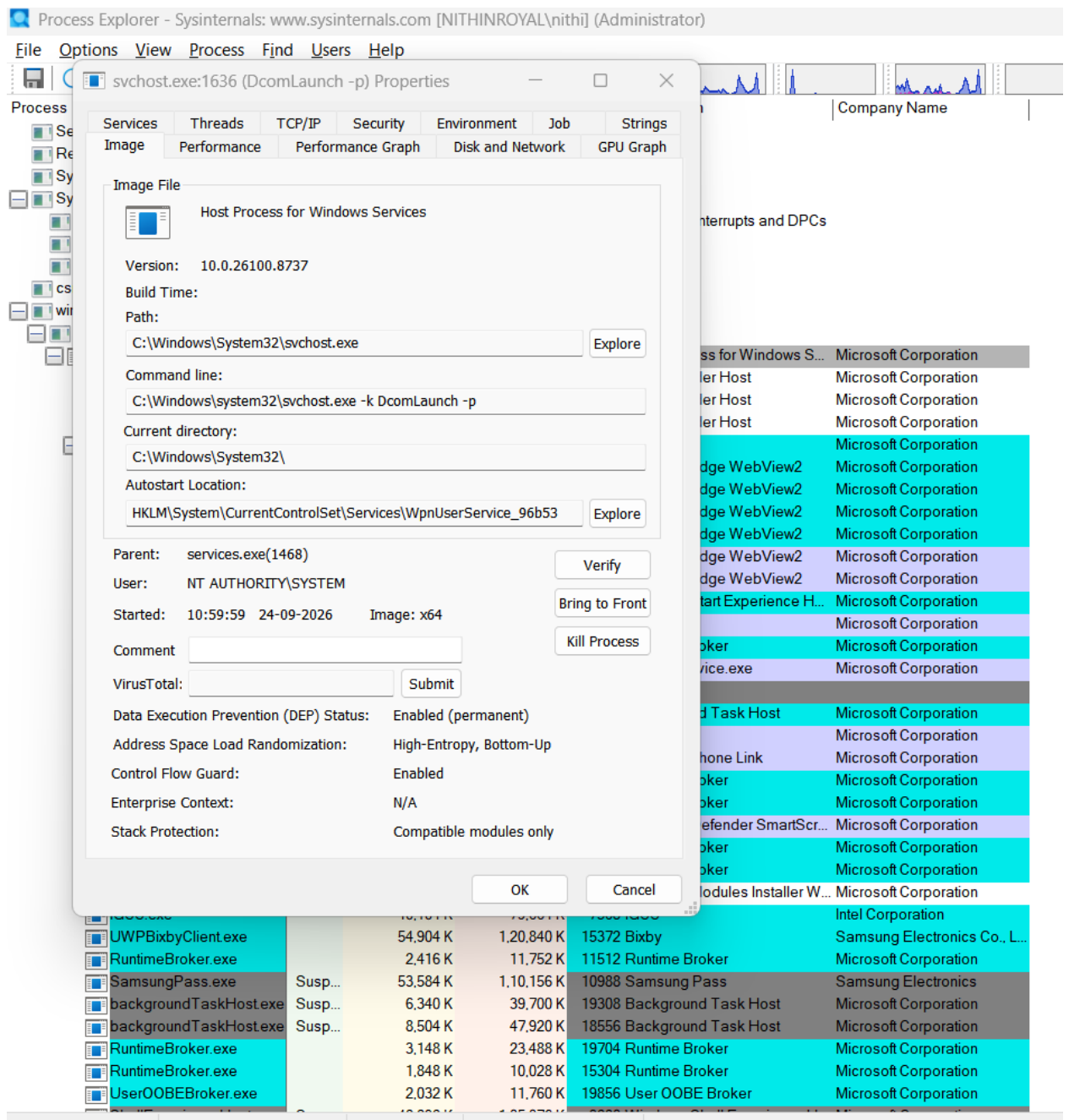
File Options View Process Find Users Help

Process	CPU	Private Bytes	Working Set	PID	Description	Company Name
Secure System		172 K	63,112 K	188		
Registry	< 0.01	19,276 K	61,252 K	232		
System Idle Process	56.89	60 K	8 K	0		
System	2.35	64 K	9,920 K	4		
Interrupts	1.68	0 K	0 K	n/a	Hardware Interrupts and DPCs	
smss.exe		1,172 K	1,672 K	868		
Memory Compression	< 0.01	568 K	2,34,444 K	3540		
csrss.exe	< 0.01	2,476 K	7,024 K	1240		
wininit.exe		1,744 K	9,160 K	1324		
services.exe	0.34	7,644 K	22,204 K	1468		
svchost.exe	0.67	14,284 K	43,396 K	1636	Host Process for Windows S...	Microsoft Corporation
WmiPrvSE.exe		8,920 K	22,508 K	6796	WMI Provider Host	Microsoft Corporation
WmiPrvSE.exe		4,756 K	13,824 K	8292	WMI Provider Host	Microsoft Corporation
WmiPrvSE.exe		3,468 K	12,788 K	10236	WMI Provider Host	Microsoft Corporation
SearchHost.exe		69,528 K	1,60,756 K	11736		Microsoft Corporation
msedgewebview2.exe		39,340 K	1,24,844 K	11272	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		3,072 K	13,140 K	13344	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		15,116 K	44,040 K	13768	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		79,616 K	71,596 K	13792	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		9,032 K	20,288 K	13916	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe	< 0.01	93,760 K	1,27,432 K	14076	Microsoft Edge WebView2	Microsoft Corporation
StartMenuExperienceHost.exe		1,03,148 K	1,83,496 K	11692	Windows Start Experience H...	Microsoft Corporation
WidgetBoard.exe		5,264 K	33,676 K	7108		Microsoft Corporation
RuntimeBroker.exe	< 0.01	9,252 K	56,788 K	5036	Runtime Broker	Microsoft Corporation
WidgetService.exe		5,160 K	26,496 K	11320	WidgetService.exe	Microsoft Corporation
SAClient.exe	Susp...	51,104 K	2,800 K	12720	SAClient	
backgroundTaskHost.exe		15,120 K	26,376 K	12832	Background Task Host	Microsoft Corporation
MicrosoftStartFeedProvi...		10,752 K	50,980 K	12496		Microsoft Corporation
PhoneExperienceHoste...	< 0.01	76,428 K	1,90,460 K	1596	Microsoft Phone Link	Microsoft Corporation
RuntimeBroker.exe		1,548 K	8,900 K	13236	Runtime Broker	Microsoft Corporation
RuntimeBroker.exe		3,108 K	19,792 K	5056	Runtime Broker	Microsoft Corporation
smartscreen.exe	< 0.01	5,628 K	29,908 K	13500	Windows Defender SmartScr...	Microsoft Corporation
RuntimeBroker.exe		3,404 K	22,496 K	9260	Runtime Broker	Microsoft Corporation
RuntimeBroker.exe		6,724 K	31,292 K	17948	Runtime Broker	Microsoft Corporation
TiWorker.exe	7.84	3,02,792 K	6,64,684 K	2956	Windows Modules Installer W...	Microsoft Corporation
IGCC.exe		40,140 K	79,688 K	7568	IGCC	Intel Corporation
UWPBixbyClient.exe		54,940 K	1,20,856 K	15372	Bixby	Samsung Electronics Co., L...
RuntimeBroker.exe		2,416 K	11,752 K	11512	Runtime Broker	Microsoft Corporation
SamsungPass.exe	Susp...	53,584 K	1,10,156 K	10988	Samsung Pass	Samsung Electronics
backgroundTaskHost.exe	Susp...	4,948 K	29,088 K	8584	Background Task Host	Microsoft Corporation
backgroundTaskHost.exe	Susp...	6,220 K	39,544 K	4288	Background Task Host	Microsoft Corporation
backgroundTaskHost.exe	Susp...	5,548 K	32,412 K	3132	Background Task Host	Microsoft Corporation
backgroundTaskHost.exe	Susp...	16,868 K	43,080 K	18516	Background Task Host	Microsoft Corporation
backgroundTaskHost.exe	Susp...	6,340 K	39,700 K	19308	Background Task Host	Microsoft Corporation

Physical Memory: 41.88% | Commit Charge: 58.64% | Processes: 390 | Physical Memory: 61.16%

Step 3: Identify Suspicious Processes

1. Look for unfamiliar or oddly named processes (e.g., xkdjeo.exe, randomname123.exe).
2. Check the Company Name and Description:
 - o Legitimate software usually shows known publishers like *Microsoft, Intel, or Adobe*.
3. Right-click the process → Properties → go to the Image tab.
4. Verify the Path of the executable file:
 - o Safe: C:\Windows\System32\
 - o Suspicious: C:\Users\<User>\AppData\Temp\ or Downloads\
5. Check for Digital Signature:
 - o Valid signature = trusted developer
 - o No signature or invalid = possibly malicious



Step 4: Analyze Process Behavior

- Observe CPU, Memory, and I/O usage columns.
- If a small or unknown process consumes excessive CPU or memory, it may be malicious.
- Right-click the process → Properties → go to the TCP/IP tab.
- Check if it communicates with unknown external IP addresses.
- Examine Handles and DLLs tabs for suspicious loaded files or libraries.

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svchost.exe:1636 (DcomLaunch -p) Properties

Services Threads TCP/IP Security Environment Job Strings
Image Performance Performance Graph Disk and Network GPU Graph

Image File

Host Process for Windows Services
(Verified) Microsoft Windows Publisher

Version: 10.0.26100.8737

Build Time:

Path: C:\Windows\System32\svchost.exe [Explore](#)

Command line: C:\Windows\system32\svchost.exe -k DcomLaunch -p

Current directory: C:\Windows\System32\

Autostart Location: HKLM\System\CurrentControlSet\Services\WpnUserService_96b53 [Explore](#)

Parent: services.exe(1468) [Verify](#)

User: NT AUTHORITY\SYSTEM [Bring to Front](#)

Started: 10:59:59 24-09-2026 Image: x64 [Kill Process](#)

Comment

VirusTotal: [Submit](#)

Data Execution Prevention (DEP) Status: Enabled (permanent)

Address Space Load Randomization: High-Entropy, Bottom-Up

Control Flow Guard: Enabled

Enterprise Context: N/A

Stack Protection: Compatible modules only

OK Cancel

Process Name	Company Name
Host Process for Windows S...	Microsoft Corporation
ler Host	Microsoft Corporation
ler Host	Microsoft Corporation
ler Host	Microsoft Corporation
dge WebView2	Microsoft Corporation
dge WebView2	Microsoft Corporation
dge WebView2	Microsoft Corporation
dge WebView2	Microsoft Corporation
dge WebView2	Microsoft Corporation
dge WebView2	Microsoft Corporation
art Experience H...	Microsoft Corporation
oker	Microsoft Corporation
vice.exe	Microsoft Corporation
d Task Host	Microsoft Corporation
hone Link	Microsoft Corporation
oker	Microsoft Corporation
oker	Microsoft Corporation
efender SmartScr...	Microsoft Corporation
oker	Microsoft Corporation
oker	Microsoft Corporation
odules Installer W...	Microsoft Corporation
UWPBixbyClient.exe	Intel Corporation
RuntimeBroker.exe	Samsung Electronics Co., L...
SamsungPass.exe	Microsoft Corporation
ShellExperienceHost.exe	Samsung Electronics
RuntimeBroker.exe	3220 Windows Shell Experience H...
WUDFHost.exe	15552 Runtime Broker
svchost.exe	Microsoft Corporation
svchost.exe	1712 Windows Driver Foundation - ...
svchost.exe	1800 Host Process for Windows S...
svchost.exe	Microsoft Corporation
svchost.exe	1848 Host Process for Windows S...
svchost.exe	Microsoft Corporation

Step 5: Verify Process Legitimacy

1. Search the process name on Google.
Example: svchost.exe vs svhost.exe (one letter missing — suspicious).

2. Visit <https://www.virustotal.com> ○ Upload the process file or search its name to verify if it's reported as malware.
3. Cross-check with **ProcessLibrary.com** or official vendor websites for authenticity.

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File Options View Process Find Users Help

svchost.exe:1636 (DcomLaunch -p) Properties

Image Services Performance Threads TCP/IP Security Performance Graph Disk and Network Environment Job GPU Graph Strings

User: NT AUTHORITY\SYSTEM
SID: S-1-5-18
Session: 0 Logon Session: 3e7
Virtualized: No Protected: No

Group	Flags
BUILTIN\Administrators	Owner
BUILTIN\Users	Mandatory
Everyone	Mandatory
LOCAL	Mandatory
Mandatory Label\System Mandatory Level	Integrity
NT AUTHORITY\Authenticated Users	Mandatory
NT AUTHORITY\LogonSessionId_0_70304	Owner
NT AUTHORITY\SYSTEM	Mandatory
NT AUTHORITY\This Organization	Mandatory
NT SERVICE\BrokerInfrastructure	Owner
NT SERVICE\DcomLaunch	Owner
NT SERVICE\DeviceInstall	Owner
NT SERVICE\LSM	Owner
NT SERVICE\PluginPlay	Owner

Group SID: n/a

Privilege	Flags
SeAssignPrimaryTokenPrivilege	Disabled
SeAuditPrivilege	Default Enabled
SeBackupPrivilege	Disabled
SeChangeNotifyPrivilege	Default Enabled
SeCreateGlobalPrivilege	Default Enabled
SeCreatePagefilePrivilege	Default Enabled
SeCreatePermanentPrivilege	Default Enabled
SeCreateSymbolicLinkPrivilege	Default Enabled

Permissions

OK Cancel

Process Name	Private Bytes	Working Set	Private Bytes	Working Set	Company Name
svchost.exe	9,620 K	19,888 K	1800	HostProcess for Windows S...	Microsoft Corporation
WUDFHost.exe	9,392 K	24,052 K	1712	Windows Driver Foundation - ...	Microsoft Corporation
WmiPrvSE.exe	4,908 K	14,964 K	6616	WMI Provider Host	Microsoft Corporation
RuntimeBroker.exe	3,752 K	26,936 K	15552	Runtime Broker	Microsoft Corporation
ShellExperienceHost.exe	47,360 K	1,10,528 K	3220	Windows Shell Experience H...	Microsoft Corporation
SamsungPass.exe	53,584 K	69,812 K	10988	Samsung Pass	Samsung Electronics
RuntimeBroker.exe	2,416 K	11,756 K	11512	Runtime Broker	Microsoft Corporation
UWPBixbyClient.exe	54,916 K	1,21,004 K	15372	Bixby	Samsung Electronics Co., L...

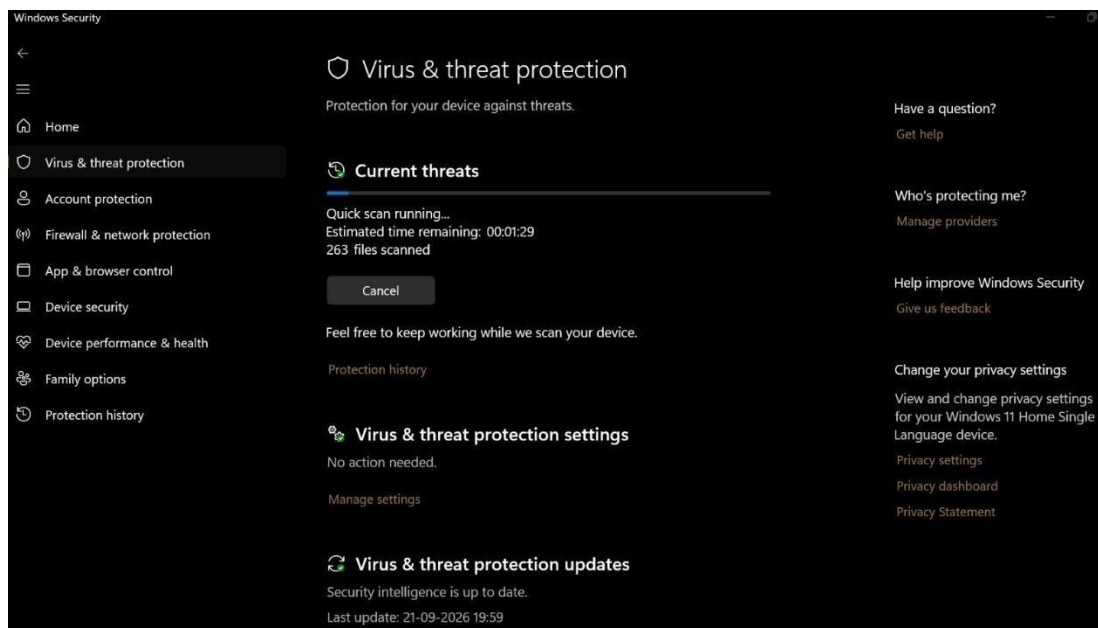
4.

Step 6: Take Appropriate Action

1. **If the process is confirmed malicious:**
 - Right-click the process → **Kill Process** to stop it.
 - Delete the corresponding executable file from its path.
2. **If unsure:** ○ Right-click → **Suspend Process** to stop it temporarily for investigation.
3. **After removal:**
 - Run a **Full System Scan** using Windows Defender or Malwarebytes to ensure no remnants remain.

You find faangpath_simple_template.pdf consuming 70% CPU.

- **Path:** C:\Users\Admin\AppData\Temp\faangpath_simple_template.pdf
- **Digital Signature:** None
- **Company Name:** Unknown
- **Network Activity:** Shows connections to unknown IPs in the TCP/IP tab
- **Online Check:** VirusTotal confirms it as a **known trojan**
- **Action Taken:** Suspended → Killed → Deleted file → Performed full antivirus scan



Rubrics

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result

Using Process Explorer, suspicious processes were successfully identified by examining their **CPU usage, path, digital signature, and network activity**.

Confirmed malicious processes were terminated and removed to maintain system integrity.
